

Smoothing: Kernels, Splines and Penalties

[Chapter 42](#) reached a natural stopping point. Piecewise polynomials can follow almost any shape, but only if someone chooses how many pieces there are and where they join, and that choice is both awkward and consequential. This chapter removes the choice. It replaces the question “how many knots?” with a question about a single continuous dial: how much roughness are we willing to pay for a closer fit? Everything else — the basis, the number of knots, the algorithm — becomes a matter of convenience once that dial exists.

Two traditions answer the question, and they meet in the middle. One fits a simple model in a moving window — a constant, a line, a cubic, weighted by distance from the point being estimated. Its dial is the width of the window, and its theory is asymptotic, a bias–variance calculation that ends in the rate $n^{-4/5}$. The other keeps a single global fit but adds a penalty on roughness; its dial is the weight on the penalty, and its theory is the algebra of [Chapter 27](#) and [Chapter 30](#): a penalized least squares problem, a linear operator, a trace that counts the degrees of freedom spent. The second tradition is the more useful one here, because a penalty carries over unchanged to several covariates ([Chapter 44](#)), to nonnormal responses and to parameters other than the mean ([Chapter 45](#)), and because it has a second reading as a normal prior, which turns the smoothing parameter into a variance ratio the data can estimate.

What you need. [Chapter 6](#) (least squares as projection, Proposition 6.3.4, and the leverage Proposition 6.8.1 that returns in [Section 43.5](#)), [Section 21.3](#) (weighted least squares, Theorem 21.3.1), [Chapter 27](#) (ridge regression Theorem 27.2.2) and [Section 30.1](#) (penalized least squares, Definition 30.1.1). From [Chapter 29](#): the leave-one-out shortcut Theorem 29.3.2, generalized cross-validation Definition 29.3.2, AIC Proposition 29.2.4 and the warning Proposition 29.5.1. From [Chapter 32](#): the linear mixed model Definition 32.2.1, Henderson’s equations Theorem 32.3.2, the BLUP Theorem 32.3.1 and REML (Definition 32.4.1, Theorem 32.4.2). [Section 1.8](#) supplies the condition number Definition 1.8.1, and [Chapter 42](#) the spline space itself (Definition 42.3.1, Theorem 42.3.1, Proposition 42.4.1). [Section 43.6](#) draws on [Section 39.4](#) for the Bayesian vocabulary (Definition 39.4.1, Proposition 39.5.1). [Section 43.4](#) solves its penalized system with the Cholesky factorization of [Chapter 10](#) (Theorem 10.2.1, Eq. 10.2.1).

Notation for this chapter. There is one covariate x taking values $x_1, \dots, x_n$ in an interval $[a, b]$, one response, and an unknown function f with $\mathbf{f} = \{f(x_1), \dots, f(x_n)\}^\top$. [Section 43.1](#) writes K for a kernel function and h for a bandwidth; from [Section 43.2](#) on, K is instead the *number of interior knots*, written $t_1 < \dots < t_K$, with κ_j reserved for the extended knot sequence of the B-spline recurrence; the kernel does not reappear except by name. $\mathbf{B}$ is a matrix of basis functions, $\boldsymbol{\gamma}$ its coefficients, $\mathbf{P}$ a penalty matrix — a local use, not the projection $\mathbf{P}$ of Chapters 4 to 33 — $\mathbf{D}$ a difference matrix and $\mathbf{S}_\lambda$ a smoother matrix. In [Section 43.4](#), $\mathbf{Q}$ and $\mathbf{R}$ are the two band matrices of the Reinsch form, a local use unrelated to the $\mathbf{R}$ of Part VII.

43.-1.1 Roadmap

- [Kernel and local polynomial regression](#)
- [B-splines](#)
- [Penalized splines](#)
- [Smoothing splines](#)
- [Choosing the smoothing parameter](#)
- [Mixed-model and Bayesian representations](#)

- [Summary and notes](#)